

Evoking Harmony via Convolution[★]

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Abstract. I show how to evoke the pitch-class content of a chord from an arbitrary source sound by convolving the source with an impulse response whose grains are one windowed sinusoid per pitch-class, across each octave of hearing range; while, at the same time, minimizing artifacts. A Csound user-defined opcode (`chord_convolver`) mixes a dry Dirac component into that response, and applies partitioned convolution once. I contrast the effect with a linear-frequency comb filter and with a generic constant- Q resonator bank, and I demonstrate musical use on a twilight field recording alongside the ruins of Château de Lagarde.

Keywords: csound, harmony, convolution, constant- Q , field recording

1 Introduction

When I was a boy my father drove with me from Salt Lake City to Los Angeles across the desert at night. Long stretches of highway were empty and straight. I lay with my head against the window and listened to the car, the road, and the wind. In that rushing noise I heard melodies and chords—tonality arising from broadband sound.

This paper reports experiments in Csound [1] that evoke such harmonies from varied sources. Conceptually, the effect convolves a source signal with an impulse response whose grains occupy only the pitch-classes of a chosen chord, realized as sinusoids across every octave within hearing range. I call the result *evoking harmony* (or harmony convolution). Materials are available online [2].

The effect sits at the boundary of timbre and tonality. A constant- Q filterbank centered on the same pitch-classes is closely related; a generic constant- Q tiling that is *not* restricted to a chord is only loosely related; a classical comb filter is even less related, because its teeth are periodic in *linear* Hz rather than in log frequency. Gabor- or Morlet-style wavelet cells, not being centered on pitch-classes, are not a good fit.

2 Motivation

There has long been a practical disjunction between “non-tonal” electroacoustic practice—often oriented toward acousmata and continuum of timbre—and tonal practice oriented toward pitch relations abstracted from actual sounds. That split goes back to the roots of computer music in the 1950s and 1960s, a period dominated by technological utopianism and musical serialism. The algorithm described here deliberately works on the boundary of timbre and tonality: it leaves the source audible while imparting a faint or strong sense of chords or chord progressions, depending on parameters and mix.

3 Mathematics

Let C be a finite set of pitch-classes. A *grain* is one causal half-cosine windowed sinusoid (pitch-class c , octave n). Its frequency response is a *kernel* $H_{c,n}$. The *impulse response* is the sum of the loudness-balanced grains plus a Dirac, then jointly ℓ_2 -normalized. For a chord C those kernels form an octave-related scaled family:

$$H_C(f) = \sum_{c \in C} \sum_n H_{c,n}(f), \quad H_{c,n}(f) \simeq H_c(2^{-n}f).$$

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In log-frequency coordinates $u = \log_2 f$, octave transposition is translation by one unit, and the total response is approximately periodic with period one octave:

$$H_C(u + 1) \simeq H_C(u).$$

Each octave therefore contains $K = |C|$ peaks whose positions within the octave are fixed by the chord. The chord acts as the unit cell of a periodic filter in log-frequency space. That is the structural difference from a comb filter, whose peaks are equally spaced in linear Hz.

Convolution with a finite impulse response incurs Gabor’s time–frequency tradeoff [3]: a shorter grain improves time resolution at the cost of frequency resolution, while a longer grain improves frequency resolution at the cost of time resolution. The duration of the grain is therefore a musical parameter. The frequency-dependent window length $T(f) = \min(T, T \cdot 440/f)$ used below distributes that tradeoff unevenly across the lattice, and deliberately so. Above a pivot at 440 Hz, $T(f) = T \cdot 440/f$ holds the number of cycles in every grain constant at $T \cdot 440$, so the treatment is constant- Q with $Q = T \cdot 440$: temporal smear falls as $1/f$, while spectral smear stays constant in the very coordinate $u = \log_2 f$ in which the chord is the unit cell. For $T = 30$ ms the measured -3 dB bandwidth is 1.68 semitones in every octave from 440 Hz upwards. Below the pivot the length is capped at T , so bandwidth is constant in Hz instead—some 43 Hz for the same T —and therefore widens in musical terms as frequency falls, to 3.4 semitones at 220 Hz and 14.6 at 55 Hz. Smearing is thus least, and the transients of the source best preserved, in the upper octaves; it is the bass that rings and blurs.

There is a further reason to hold the grains short, and it is perceptual rather than spectral. The aim is that the harmony seem latent in the source and evoked from it, not added to it. Convolution serves that aim by construction, since $Y(f) = H(f)X(f)$ vanishes wherever the source has no energy: convolution can only expose pitch content that the source already contains, never introduce any. What can still betray the effect is time. Common onset binds concurrent sounds into a single event, but energy persisting after the source has fallen silent would be a cue for a separate, struck resonator – not a property of the source. The causal window and the cap on $T(f)$ both serve auditory fusion in this sense, by keeping the harmonic response within the duration of the source. On a 50 ms impulse response driven by a 3 Hz impulse train, the design used here leaves under 3 % of each period’s energy outside the interval the dry source occupies, whereas an uncapped or symmetrically windowed grain displaces a third or more. On sustained material the tail is never exposed and longer grains cost nothing, which makes T properly a function of the transient density of the source rather than a constant of the method.

4 Method

4.1 Implementation

The Csound user-defined opcode is `chord_convolver`. It builds one impulse response h , and applies partitioned convolution once via `ftconv` [4]. The design choices are:

1. For each MIDI note whose pitch-class is selected, include that frequency if it lies between 20 Hz and 0.95 of Nyquist.
2. Give partial f a causal half-cosine grain of length

$$T(f) = \min(T, T \cdot 440/f),$$

keeping ringing equal across higher partials. A full Hann peaking mid-window would delay the audible onset of the grain, loosening its fusion with the attack of the source; the half-cosine starts at 1 and fades to 0.

3. Scale each partial by $A(f) \propto T(f)^{-1/2}$, so that all partials carry equal energy despite their differing lengths (Section 4.2).
4. Weight each partial by inverse IEC A-weighting (unity at 1 kHz), with a rumble shelf that evaluates R_A no lower than 150 Hz and a boost cap of +6 dB, so that low wind and rumble do not dominate.
5. Scale grains by an impulse gain, average across partials, add a Dirac component at sample 0 (dry path inside the IR), apply a joint ℓ_2 normalization to the whole IR, and convolve.

4.2 Levelling Octaves

Because $T(f)$ shrinks with frequency, partials of equal amplitude are not equally present. Levelling them proceeds in two stages: first equalize the physical contribution of each partial, then convert that into equal perceived loudness.

For the first stage the relevant quantity is grain energy, not amplitude and not the window integral. The sources of interest are broadband, so for excitation of power spectral density S_0 the output power contributed by partial k is, by Parseval's theorem,

$$P_k = S_0 \int |G_k(f)|^2 df = S_0 \int g_k(t)^2 dt = S_0 E_k,$$

so equal presence means equal E_k , giving $A(f) \propto T(f)^{-1/2}$. A constant factor then restores the total grain energy, so that the impulse and Dirac gains retain their meaning independently of this choice.

Loudness tracks that power rather than spectral density because every grain is narrower than one auditory filter: measured against the auditory equivalent rectangular bandwidth (ERB) of [5], the widest grain in the lattice occupies 0.74 of a critical band and the rest fall below 0.5. This is why the window integral $A(f)T(f)$, which fixes the peak transfer gain $|H(f_k)|$, is the wrong target: as $T(f)$ shortens, the grain's bandwidth $\approx 1/T(f)$ widens, and holding peak gain flat across a widening band yields rising power. Normalizing the integral instead of the energy measurably tilts the lattice bright by some 8 dB from the bass to the top octave.

The second stage is the inverse A-weighting of item 4 above, which converts equal power per partial into approximately equal loudness per partial. Measuring the realized impulse response confirms the two stages are cleanly separated: the per-partial band powers follow the A-weighting curve to within 0.02 dB across the lattice, so nothing but the intended perceptual weighting remains.

Equal loudness *per partial* is not the same as equal loudness *per critical band*. Since an ERB spans some 27 semitones at 20 Hz but only 1.9 semitones above 4 kHz, the low octaves crowd several chord tones into one auditory filter, where their powers sum. Dividing out that crowding is available as an option, and attenuates the lowest octaves by 4 to 6 dB; the recordings here do not use it, keeping each lattice tone reading equally rather than evening out the harmony as a whole.

Pseudocode:

```
chord_convolver(x, T, g_imp, g_dirac, compensation, PCs...):
    T      = clamp(abs(T), 0.01, 1.5)
    n_max  = T * 440
    N      = max(2, round(T * sr))
    // scale degrees reduce to pitch-classes; negatives are unused
    PCs    = { p mod 12 : p in PCs, p >= 0 }
    freqs  = []; weights = []; Tf = []; amp = []
    for midi = 0..127:
        if (midi mod 12) not in PCs: continue
        f = cpsmidinn(midi)
        if f < 20 or f > 0.95*Nyquist: continue
        fp = max(f, 150)           // rumble shelf
        w  = R_A(1000) / R_A(fp)   // inverse A-weight
        w  = min(w, 10^(6/20))     // cap boost +6 dB
        append f to freqs; append w to weights
        append min(T, n_max / f) to Tf
    M = length(freqs)
    p = [0, 0.5, 1, 0.5][compensation] // 3 also * D_k^-1/2
    for i = 0..M-1: amp[i] = (T / Tf[i])^p
    c = sqrt(sum(Tf) / sum(amp^2 * Tf)) // hold total energy
    for i = 0..M-1: amp[i] = c * amp[i]
    h = zeros(N)
    for n = 0..N-1:
        t = n / sr; s = 0
        for i = 0..M-1:
            if t < Tf[i]:
```

```

    env = 0.5 + 0.5*cos(pi*t/Tf[i])    // causal, 1 -> 0
    s = s + weights[i]*amp[i]*env*sin(2*pi*freqs[i]*t)
    h[n] = (s / M) * g_imp
    h[0] = h[0] + g_dirac
    h     = h / L2norm(h)
    return ftconv(x, h)

```

Earlier experiments compared IIR resonator banks and multi-`ftconv` variants; the joint IR of `chord_convolver` is the implementation used in the comparison spectrograms and the sound walk below.

4.3 What It Is Not

A *constant-Q* bank with centers restricted to the chord’s pitch-classes is a near neighbour: same log lattice, different dynamics (ongoing filtering versus convolution with a finite impulse response). A generic *constant-Q* tiling spaced every few semitones (not chord-restricted) emphasizes a regular pitch grid without chord identity. A *comb* filter with delay $1/f_0$ emphasizes harmonics of f_0 in linear Hz; on a linear-frequency spectrogram its teeth are equally spaced, whereas the chord convolver (and a chord-centered *constant-Q* bank) show an octave-periodic pattern.

5 Results

5.1 Click and Noise Comparison

Figure 1 compares treatments of a Dirac-like click and of white noise: dry source; comb with $f_0 = 200$ Hz; a sparse *constant-Q* bank (centers every three semitones); and `chord_convolver` with pitch-classes 0, 4, 7, 11 (C–E–G–B, CM7). Segments are loudness-matched for listening, to within 1.1 dB A-weighted within each source group. The comb shows equal spacing in linear Hz; the *constant-Q* bank shows log spacing without chord restriction; the convolver shows a sparser chordal lattice, which under equal-energy normalization stays articulated to the top of the audible range rather than fading out above the low mids.

5.2 Sound Walk

At twilight I walked up from my village in France and alongside the ruins of Château de Lagarde and used a Tascam DR-40 to record my footsteps in fallen leaves, voices from the village, cars, wind in trees, church bells, and jackdaws. A Csound instrument applies `chord_convolver` under score control of onset, duration, fades, grain duration T , wet and dry gains, compensation type, and any number of pitch-classes.

The first section of this score is (compensation 1 = equal energy; pitch-classes are MIDI keys reduced modulo 12):

; instr	onset	duration	fadein	fadeout	T	wet	dry	comp	pitch-classes				
i "evoke"	0.000	29.266	8.00	1.00	0.03	0.30	0.6	1	0	4	7	11	14
i "evoke"	29.266	15.663	1.00	1.00	0.04	0.10	0.6	1	2	5	9	12	14
i "evoke"	44.929	52.171	1.00	1.00	0.06	0.15	0.7	1	5	7	9	14	
i "evoke"	97.100	26.208	1.00	1.00	0.05	0.10	0.5	1	2	5	9	12	4
; Three bell strikes													
i "evoke"	123.308	1.832	0.05	0.05	0.05	0.10	0.9	1	7	10	13	15	
i "evoke"	125.140	1.782	0.05	0.05	0.05	0.20	0.9	1	3	6	10	13	15
i "evoke"	126.922	22.078	0.05	10.00	0.02	0.10	0.8	1	0	4	7	11	14

Effects can be subtle or obvious. Instructive moments include a chord progression within voices, bells with a subtle effect, bells again with a stronger effect and voices following, and “Bonjour” with voices following. Audio and further materials are in the repository [6,2].

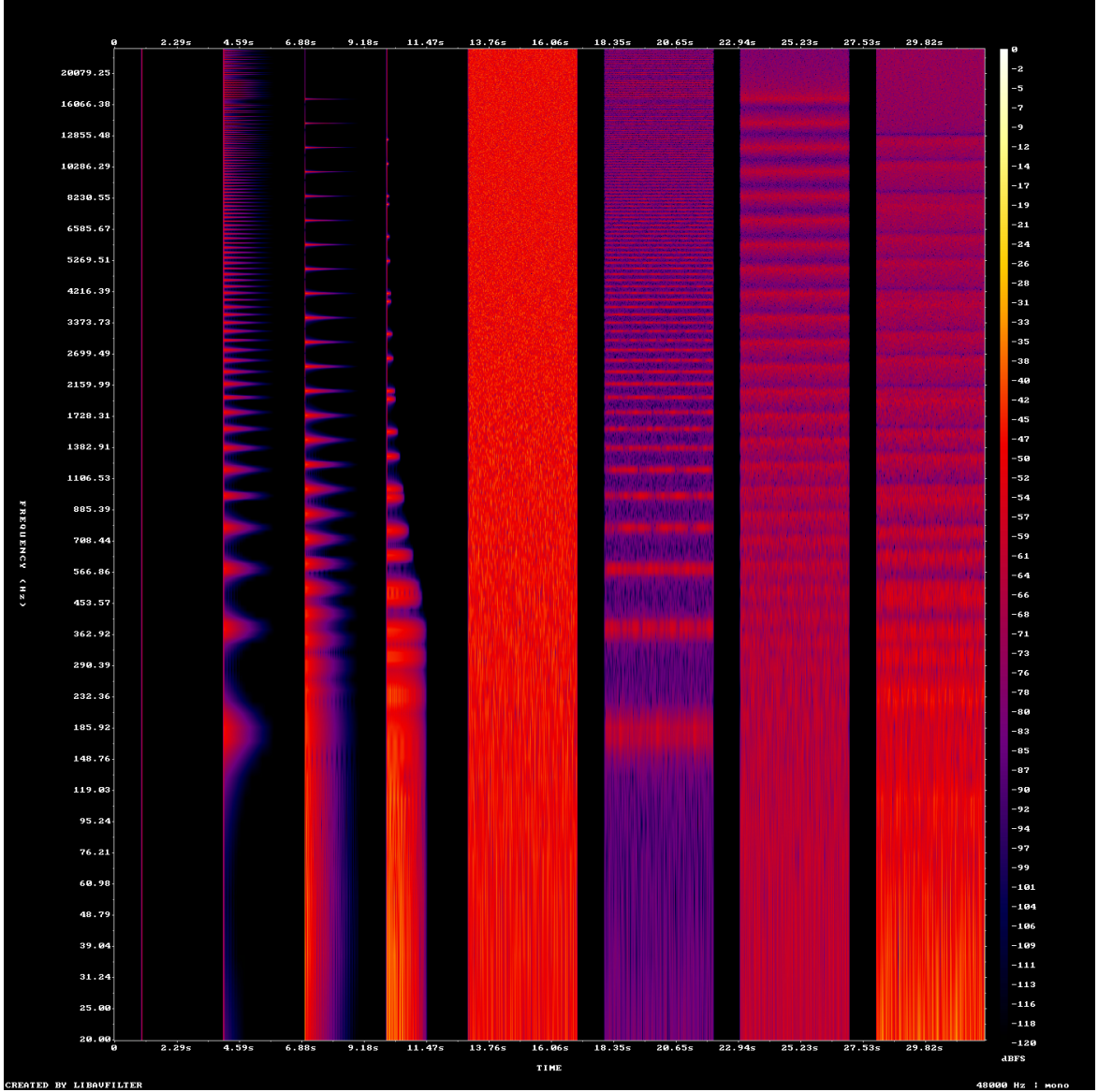


Fig. 1. Log-frequency spectrogram (left to right): dry click; click comb; click CQT; click evoke CM7; then the same four treatments of noise.

6 Discussion

Evoking harmony is related to comb filtering and to constant- Q analysis, but it is not the same as either. The lattice is logarithmic and chord-specific, and the preferred realization is causal grains of short to medium duration plus a dry Dirac in one impulse response. It works best when harmonic sensation remains subordinate to the source timbre—a chordal coloration in broadband or mixed field sound. Wetter settings and pitched sources can produce a clearer “harmonic shadow,” ringing, or even increasingly abstract rhythmic and harmonic textures. Parameter choices (T , wet and dry gains, compensation type, pitch-classes, A-weight shelf) are all audible, and all musical.

The levelling of Section 4.2 is a compositional choice as well as a technical one. Equalizing amplitude, energy, or window integral spans roughly 17 dB of spectral tilt across the lattice, from a dark balance in which the upper octaves vanish to a glassy one in which they dominate. The recordings here use equal energy, so that each pitch-class is equally present. Equalizing critical bands rather than partials is a further, audibly distinct option.

References

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